

Joint Somatic Phasing and Purity Estimation Incorporating Loss of Heterozygosity in Tumor-Only Long-Read Sequencing

Yao-Ting Huang and Zhen-Yu Chen

Department of Computer Science and Information Engineering, National
Chung Cheng University.

Contributing authors: ythuang@cs.ccu.edu.tw;

Abstract

The characterization of somatic haplotypes from tumor-only long-read sequencing data is critical for precision oncology but is confounded by interdependent challenges, including large-scale Loss of Heterozygosity (LOH) and unknown tumor purity. Existing analytical approaches often fail to model these biological phenomena jointly, leading to fragmented and inaccurate genomic interpretations. This thesis introduces LongPhase-TO, a novel, integrated bioinformatics pipeline designed to overcome these limitations. Using a unified statistical framework, LongPhase-TO simultaneously performs somatic phasing, detects chromosome-scale LOH, refines somatic variant calls, and estimates tumor purity from a single tumor specimen. Validation on well-characterized cancer cell lines and in-silico mixtures demonstrates that LongPhase-TO achieves superior phasing contiguity compared to standard germline phasers and accurately identifies LOH events with high concordance to gold-standard microarray data. Furthermore, the pipeline significantly enhances the precision of upstream somatic variant callers by leveraging haplotype information as a post-processing filter. A key capability is its highly accurate prediction of tumor purity derived from haplotype imbalance signals, obviating the need for a matched normal sample and outperforming established methods. Qualitative visualizations confirm its power to resolve the specific parental origin of somatic mutations, even when co-occurring with large structural events. By providing a comprehensive, haplotype-resolved genomic profile from a single tumor sample, LongPhase-TO establishes a powerful new methodology for cancer genomics, advancing the utility of long-read sequencing for both fundamental research and clinical applications where only tumor material is available.

Keywords: Long-read sequencing, Tumor-only, Somatic phasing, Loss of Heterozygosity (LOH), Tumor purity estimation

Introduction

Cancer is fundamentally a disease of the genome, initiated and propelled by the accumulation of somatic mutations [1–3]. These acquired genetic alterations, which arise in a specific cell during an individual’s lifetime, are distinct from the germline variants inherited from one’s parents and present in every cell of the body (Supplementary Fig. 8). This distinction is paramount in oncology, as identifying the unique landscape of somatic mutations within a tumor is essential for elucidating its biological behavior, diagnosing its subtype, and selecting targeted therapies. While the advent of high-throughput sequencing has transformed cancer research by enabling the comprehensive cataloging of these mutations, a mere inventory of variants is insufficient. To fully comprehend their functional impact, it is necessary to resolve their chromosomal context, a process known as haplotype phasing.

Haplotype phasing is the computational process of assigning variants to their chromosome of origin, thereby reconstructing the distinct genetic sequences inherited from each parent. This resolution is critical because the functional consequence of multiple mutations is contingent upon whether they occur on the same chromosome copy (in cis) or on different copies (in trans) [4, 5]. The emergence of long-read sequencing technologies has significantly advanced phasing capabilities [6, 7]. By generating reads that span tens of thousands of base pairs, these platforms can physically link numerous heterozygous variants on a single DNA molecule, facilitating the reconstruction of long, contiguous haplotype blocks. This technological leap has largely resolved the challenge of germline phasing, which aims to reconstruct the two stable, inherited haplotypes of an individual (Supplementary Fig. 9).

In the domain of germline variant analysis, sophisticated computational methods have been developed to leverage the power of long-read data. A key challenge in this field is distinguishing true genetic variants from the inherent noise of sequencing errors. State-of-the-art tools, such as DeepVariant and Clair3, employ deep learning models that analyze patterns in aligned sequencing reads to render highly accurate predictions, effectively separating the biological signal from technological artifacts [8–10]. These methods first phase the reads into two haplotype-specific groups and subsequently supply this partitioned data to a neural network, demonstrating that phasing is a foundational step for achieving high-fidelity variant calling. This principle of utilizing phase information to enhance signal has been successfully extended to cancer genomics.

The standard approach for identifying somatic mutations is based on comparative analysis of matched tumor and normal samples from the same patient [11–13]. This approach permits the direct subtraction of the germline background, thereby isolating variants unique to the tumor. Modern somatic variant callers designed for long-read

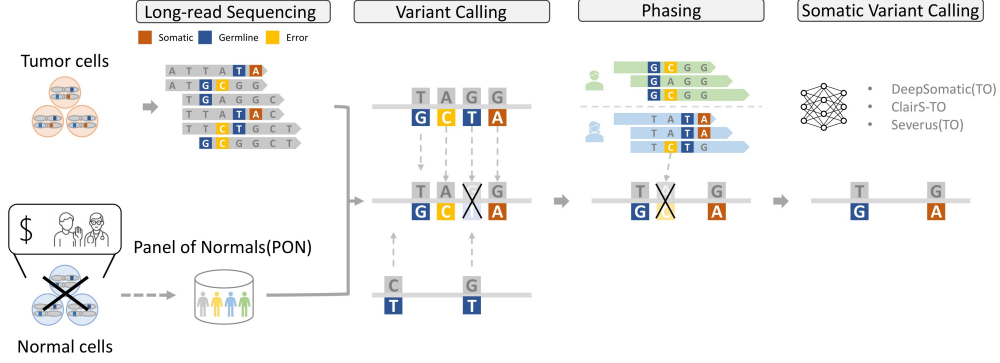


Fig. 1 Schematic overview of tumor-only somatic variant calling workflow. In the absence of matched normal samples, long-read sequencing of tumor DNA generates reads containing a mixture of somatic variants (orange), germline variants (blue), and sequencing errors (yellow). Candidate variants are first filtered against PONs to remove common germline variants and recurrent technical artifacts. Subsequently, phasing reconstructs haplotypes to distinguish true somatic mutations from error signals. Finally, tumor-only callers such as DeepSomatic(TO), ClairS-TO, and Severus(TO) integrate phasing information to output high-confidence somatic variant calls. Although slightly less accurate than analyses using paired tumor-normal samples, this workflow provides reliable variant detection in tumor-only contexts.

data, including DeepSomatic, ClairS, and Severus, have adopted a phase-centric strategy [14–16]. By first phasing the reads from the tumor sample, they concentrate the signal of a somatic mutation, which typically arises on a single haplotype, into one partition of the data. This segregation markedly improves the signal-to-noise ratio, enabling machine learning models to more readily distinguish a low-frequency somatic variant from randomly distributed sequencing errors.

In many clinical and research settings, however, a matched normal sample is unavailable, which necessitates a tumor-only workflow. This scenario presents a significant analytical challenge: differentiating a true, low-frequency somatic mutation from a rare germline variant without a constitutional baseline from the same individual [17–19]. The standard mitigation strategy involves filtering candidate variants against a Panel of Normals (PON), a database of common germline variants and recurrent technical artifacts compiled from a large cohort of healthy individuals [11]. Although this method effectively eliminates many non-somatic variants, the substantial inter-individual variability means that no database can comprehensively capture all mutations. Consequently, specialized tumor-only callers such as DeepSomatic(TO), ClairS-TO and Severus(TO) continue to rely on phasing to enhance the somatic signal, underscoring its indispensable role even in the most challenging data contexts [14, 16, 20] (Fig. 1).

Despite these advances, a critical gap persists in the field: the inability of current tools to perform true somatic phasing. Existing methods excel at germline phasing within a tumor sample, meaning that they can separate reads according to the two inherited parental haplotypes, as exemplified by tools such as WhatsHap, HapCut2, and LongPhase [21–23]. However, they are incapable of reconstructing the novel, complex somatic haplotypes that are the products of a tumor’s evolution. Cancer is a

dynamic process characterized by clonal evolution, where successive rounds of mutation and selection give rise to a heterogeneous population of subclones, each possessing a unique genetic identity [24]. Reconstructing the specific somatic haplotypes of these clones is essential for understanding co-mutation patterns, tracing tumor lineage, and deciphering the cooperative mechanisms that drive malignancy. This represents a frontier in cancer genomics that current methodologies have yet to cross.

The path toward accurate somatic phasing is obstructed by two major biological and computational hurdles inherent to cancer genomes. The first is the prevalence of large-scale structural changes, particularly Loss of Heterozygosity (LOH), where vast regions of a chromosome lose one of the two parental copies [25–27]. Such events disrupt the continuity of heterozygous markers upon which standard algorithms depend, leading to algorithmic failure and the production of fragmented, biologically inaccurate phase blocks. Achieving contiguous, chromosome-scale phasing requires a framework that can explicitly model and navigate these genomic discontinuities. The second challenge is tumor purity; clinical tumor samples are almost invariably a mixture of cancer cells and contaminating normal cells [28–30]. This cellular heterogeneity introduces conflicting haplotype signals into the sequencing data, which confounds phasing algorithms and prevents the reliable determination of the haplotype of somatic origin [31].

This thesis aims to overcome these fundamental limitations by developing LongPhase-TO, an integrated computational framework for the comprehensive analysis of tumor-only long-read sequencing data. The primary objective is to advance beyond conventional variant calling to achieve true somatic phasing. Through the development of novel algorithms, this method is designed to reconstruct the evolutionary lineage of somatic haplotypes that define a tumor’s clonal architecture; to robustly handle chromosome-scale LOH, ensuring contiguous and accurate phasing across complex genomic rearrangements; and to jointly estimate tumor purity, thereby deconvolving the mixed signals from tumor and normal cells to prevent phasing errors. By addressing these challenges within a unified model, this work seeks to provide a more precise and complete reconstruction of the cancer genome, enabling deeper insights into the evolutionary dynamics of tumorigenesis.

Results

Overview of the LongPhase-TO framework

LongPhase-TO is an extension of the LongPhase algorithm, specifically designed for somatic haplotype phasing in tumor-only sequencing data. (Fig. 2) illustrates the workflow and overall architecture of LongPhase-TO. The framework achieves somatic haplotype phasing through four core modules, including haplotype-aware variant re-calling, chromosome-scale detection of loss of heterozygosity (LOH), somatic phasing, and tumor purity estimation. Each of these modules operates based on haplotype structures to enable more accurate and comprehensive somatic phasing results. Beyond haplotype reconstruction, LongPhase-TO also reports refined variant calls, identified LOH regions, and tumor purity estimates, providing a comprehensive and integrated interpretation of the somatic genome.

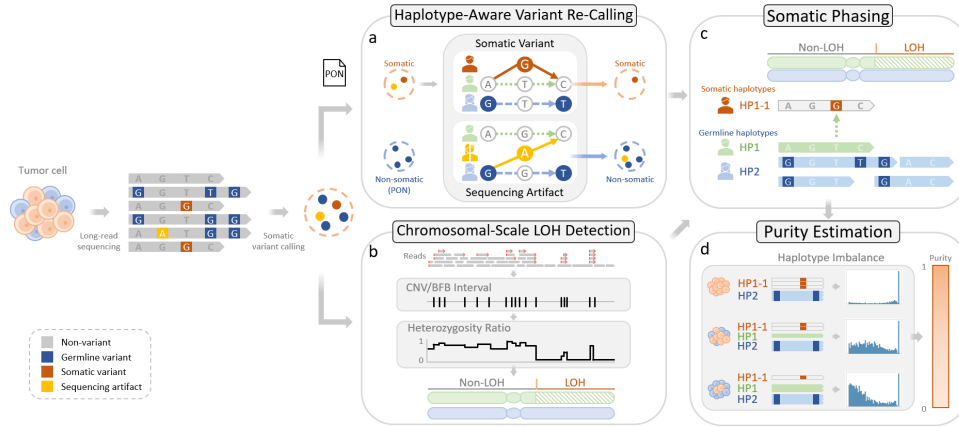


Fig. 2 Overview of the LongPhase-TO somatic haplotype phasing framework. (a) Haplotype-aware variant re-calling refines initial somatic variant calls by integrating PONs and haplotype linkage information, distinguishing true somatic variants (orange) from sequencing artifacts (yellow) and germline variants (blue). (b) Chromosome-scale LOH detection identifies large-scale allelic imbalances by analyzing CNV/BFB intervals and heterozygosity ratios to distinguish LOH from non-LOH regions. (c) Somatic phasing assigns somatic variants to their originating germline haplotypes (HP1, HP2) and reconstructs derived somatic haplotypes (HP1-1, HP2-1), spanning LOH regions. (d) Tumor purity estimation quantifies tumor content based on haplotype imbalance, where deviation from balanced allelic representation reflects increasing tumor purity.

Superiority of Somatic-Aware Phasing

To assess phasing performance, we analyzed several benchmark cancer cell line datasets, including HCC1395 from Zheng *et al.*[15] and two COLO829 datasets from the New York Genome Center (NYGC) and Epi2me (Oxford Nanopore Technologies). We further incorporated five additional datasets, namely HCC1395, HCC1937, HCC1954, H1437 and H2009 from Park *et al.*[14], which are widely used for benchmarking somatic variant callers.

All samples were sequenced using the Oxford Nanopore platform. Sequencing reads were downsampled to a coverage depth of 50×; if the original coverage was below this threshold, the full dataset was used for evaluation (see Supplementary Note [Downsampling](#) for details on sampling strategy and coverage normalization).

As all phasing tools depend on variant calls as input, we first performed variant calling using the ClairS-TO pipeline (Supplementary Note [ClairS-TO](#) for software configuration and parameter settings). Using variant sites generated by ClairS-TO, we compared the performance of four phasing tools including WhatsHap, LongPhase, HapCUT2 and the proposed somatic-aware phaser LongPhase-TO across eight cancer datasets.

The phased ratio of LongPhase-TO ranged from 0.57 to 0.71, whereas the other methods achieved 0.40–0.65. In six out of eight samples, LongPhase-TO outperformed the next-best method by an average of 0.10–0.30, and by up to 0.71 in HCC1937_UCSC (0.71 versus ~0.40), with comparable performance observed only in HCC1395_HKU and HCC1954_UCSC (Fig. 3a). Notably, both HCC1395_HKU

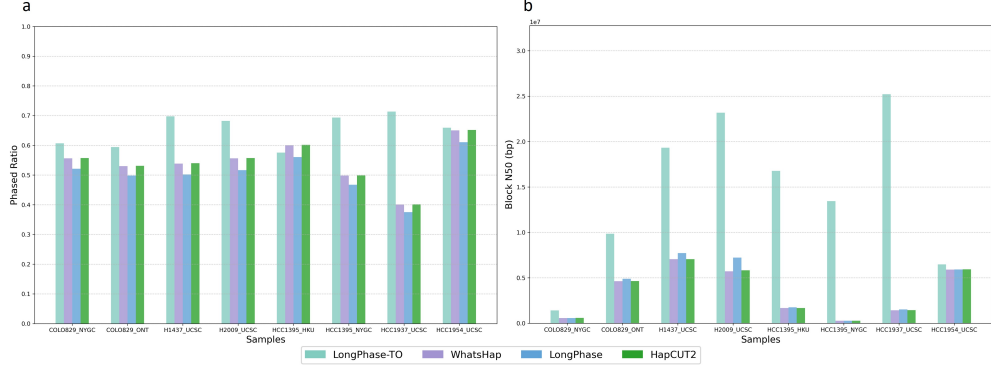


Fig. 3 Performance comparison of LongPhase-TO (somatic-aware) and LongPhase (germline) across eight cancer cell line datasets. Bars represent four pipelines: LongPhase-TO with ClairS-TO-ssrs calls (teal) and ClairS-TO-ss calls (light blue), and LongPhase with the same two variant sets (green and dark blue). (a) Phased Ratio: LongPhase-TO consistently achieves higher phasing completeness across all samples and input variant sets, typically in the range of 0.55–0.62. (b) Block N50: LongPhase-TO produces substantially longer phase blocks—often spanning several megabases and, in some datasets, reaching 20–25 Mb—whereas LongPhase results in shorter N50 values. These results indicate that incorporating homozygous sites through somatic-aware phasing improves both completeness and contiguity in tumor-only sequencing data.

and HCC1954_UCSC also exhibited higher Block N50 values than all other methods. In HCC1937_UCSC, H2009_UCSC, H1437_UCSC, HCC1395_HKU, and HCC1395_NYGC, the Block N50 exceeded 10 million base pairs (Mbp), representing a 2–10-fold improvement in phase continuity compared with other methods, which achieved a maximum of 7.7 Mbp (Fig. 3b).

Application and Validation of Phasing for LOH Detection

Having established the superior phasing capabilities of LongPhase-TO, we next evaluated its direct application in detecting Loss of Heterozygosity (LOH), a common and significant event in cancer [25, 32]. This evaluation involved two stages: first, optimizing the input for the LOH detection pipeline, and second, validating the final pipeline against a gold-standard orthogonal technology.

Impact of Input Somatic Variant Calls on LOH Detection

To evaluate the feasibility of detecting chromosome-scale loss of heterozygosity (LOH) in tumor cell lines, three distinct variant-calling inputs were analyzed using LongPhase-TO across multiple cancer samples. The results revealed consistent overall trends among datasets, while substantial inter-sample variation was observed (Figure 4).

In terms of LOH ratio, the COLO sample exhibited LOH in approximately 30% of the genome. Comparable patterns were observed in other cell lines, with HCC1395 displaying nearly half of its genome under LOH. Regarding the largest LOH segments, the COLO sample contained a segment of approximately 110 Mb, corresponding to

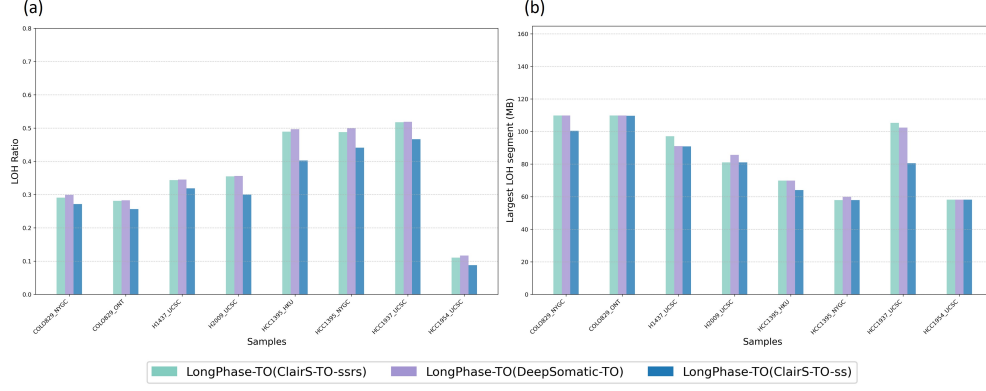


Fig. 4 Comparison of LOH ratio across multiple cancer samples using three different LongPhase-TO configurations. Panel (a) shows LOH ratio distributions for each sample, while panel (b) presents the size of the largest LOH segment in megabases. Consistent trends were observed across the different variant-calling inputs, although substantial variation was evident among individual cell lines.

nearly half the length of chromosome 1; other cell lines also harbored similarly large segments.

In contrast, the HCC1954_UCSC sample showed lower LOH ratios and shorter LOH segments relative to the other cell lines. Overall, these findings demonstrate pronounced variability in LOH across tumor cell lines and underscore the effectiveness of LongPhase-TO in detecting chromosome-scale genomic alterations.

Validation Against Microarray-Based LOH Profiles

To validate the accuracy of our sequencing-based LOH detection method, we performed a head-to-head comparison against results from the Affymetrix CytoScan HD array, a widely used and well-established microarray platform for detecting copy number alterations and LOH [33, 34]. This analysis was conducted on the HCC1395_HKU cancer cell line, with LOH calls from our LongPhase-TO pipeline compared against the CytoScan HD reference across the entire genome (Figure 5).

A genome-wide visualization, presenting each chromosome from 1 to 22, X, and Y, reveals a remarkably high level of concordance between the two methods. For each chromosome, the LOH regions detected by the LongPhase-TO pipeline (represented by blue bars in the corresponding figure) show near-perfect overlap with the LOH regions detected by the CytoScan HD array (purple bars). This strong agreement is evident across numerous chromosomes, including large-scale events on chromosomes 1, 3, 5, 8, 10, 11, 12, 17, and X. The long, contiguous LOH blocks identified by LongPhase-TO are a direct manifestation of its superior long-range phasing capability. This visual validation provides compelling evidence that the LongPhase-TO-based methodology accurately identifies known, large-scale LOH events, confirming its efficacy and reliability as a tool for LOH analysis from tumor-only long-read sequencing data.

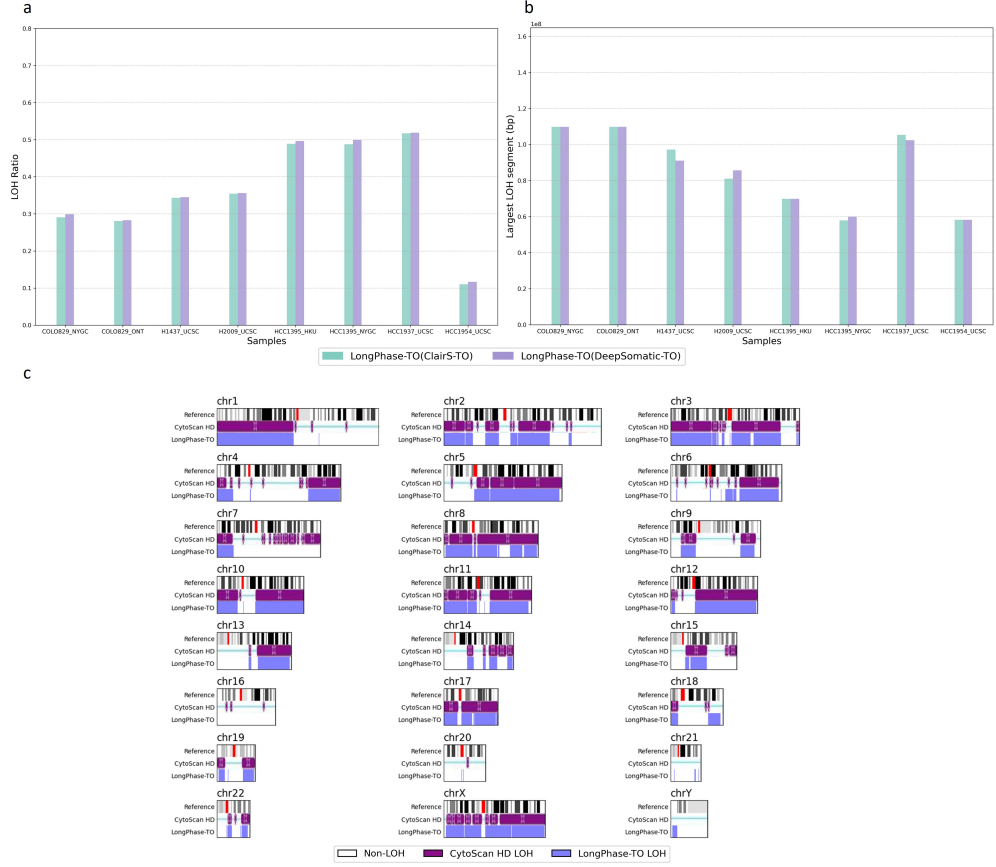


Fig. 5 Genome-wide comparison of LOH profiles between the sequencing-based LongPhase-TO pipeline (blue) and the Affymetrix CytoScan HD array (purple) in the HCC1395_HKU cancer cell line. Across all chromosomes, a high degree of concordance is observed between the two platforms, indicating that LongPhase-TO reliably recapitulates microarray-based LOH detection at chromosome scale.

Refining Somatic Variant Calls with Phasing Information

Beyond LOH detection, haplotype information can be leveraged to refine the accuracy of upstream somatic variant calling [35]. We investigated the impact of integrating LongPhase-TO as a post-processing filter for somatic single nucleotide variant (SNV) calls. This was assessed by comparing the performance of base callers with and without the LongPhase-TO refinement step, using models trained on both real and simulated data.

Performance on Models Trained with Real Data

We first evaluated the impact of LongPhase-TO on two state-of-the-art somatic callers, ClairS-TO-ssrs and DeepSomatic-TO, using models trained on real sequencing data.

The performance of four pipelines—the two base callers alone and the two in combination with LongPhase-TO—was benchmarked across eight cancer cell line datasets at varying levels of simulated tumor purity (20% to 100%). Performance was measured using the F1-score, precision, and recall (Figure 6).

The results, presented in a comprehensive grid of performance plots, show that applying LongPhase-TO as a filter has a distinct and significant effect on each base caller. For ClairS-TO-ssrs, the addition of LongPhase-TO consistently improves both precision and the overall F1-score across nearly all datasets and purity levels. This indicates that LongPhase-TO acts as an effective filter, removing false positives from the ClairS-TO-ssrs call set and thereby increasing its overall accuracy. In the case of DeepSomatic-TO, the baseline caller exhibited very high recall but insufficient precision. Incorporating LongPhase-TO markedly enhanced its precision, thereby improving the reliability of the resulting call set. Together, these observations demonstrate that LongPhase-TO can systematically refine outputs from distinct somatic callers, albeit with caller-specific trade-offs.

For the HCC1937 dataset, precision exhibited a marked improvement of approximately 20% across tumor purity levels ranging from 0.2 to 0.8, while recall showed modest reductions that varied with purity. Nevertheless, the overall F1-score consistently increased, reflecting the net gain in accuracy. Similar patterns were observed across the other cancer cell line datasets: precision was substantially improved in nearly all cases, whereas recall showed varying degrees of reduction. Importantly, when integrated into the F1-score, these trade-offs consistently resulted in an overall enhancement of performance.

As expected, the performance of all tested pipelines generally improved with increasing tumor purity, as the higher proportion of tumor DNA makes somatic variants more readily distinguishable from the germline background. These results validate LongPhase-TO as a powerful refinement tool that can enhance the accuracy of somatic variant calling.

Accurate Tumor Purity Estimation from Tumor-Only Long-Read Data

An accurate estimate of tumor purity is a critical parameter for interpreting cancer genomics data [36, 37]. We assessed the capability of our LongPhase-TO-based pipelines to predict tumor purity directly from tumor-only long-read sequencing data. The purity estimates from three LongPhase-TO pipeline variants were compared against those from ASCAT, a widely used tool that requires matched tumor–normal data, using simulated purity as the ground truth (Figure 7).

The evaluation, conducted across eight cancer cell line datasets, demonstrates the strong performance of the LongPhase-TO-based methods. In a series of scatter plots comparing estimated purity against simulated purity, the predictions from all three LongPhase-TO pipelines closely tracked the ideal prediction line ($y = x$) across the full range of purity levels (20% to 100%). This high accuracy was largely consistent across datasets and appeared independent of the specific upstream variant caller used (ClairS-TO-ssrs, ClairS-TO-ss, or DeepSomatic-TO), suggesting the robustness of the purity estimation algorithm within LongPhase-TO itself. Minor deviations from the

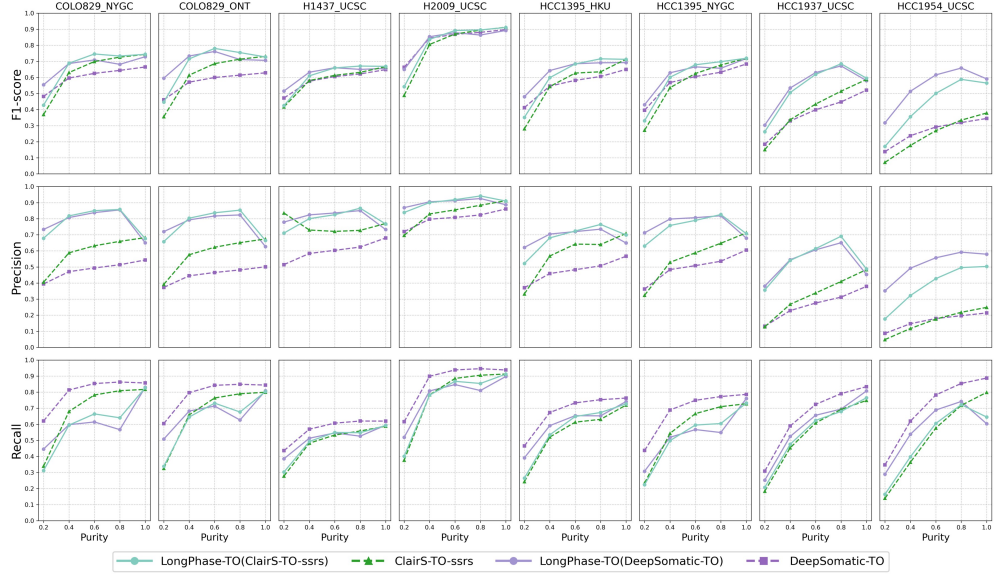


Fig. 6 Performance comparison of ClairS-TO-ssrs and DeepSomatic-TO, with and without LongPhase-TO, across eight cancer cell line datasets under varying tumor purities (20%–100%). Incorporation of LongPhase-TO consistently increased precision and F1-score, with modest reductions in recall. Overall, LongPhase-TO improved the accuracy of somatic variant detection, particularly at intermediate tumor purity levels.

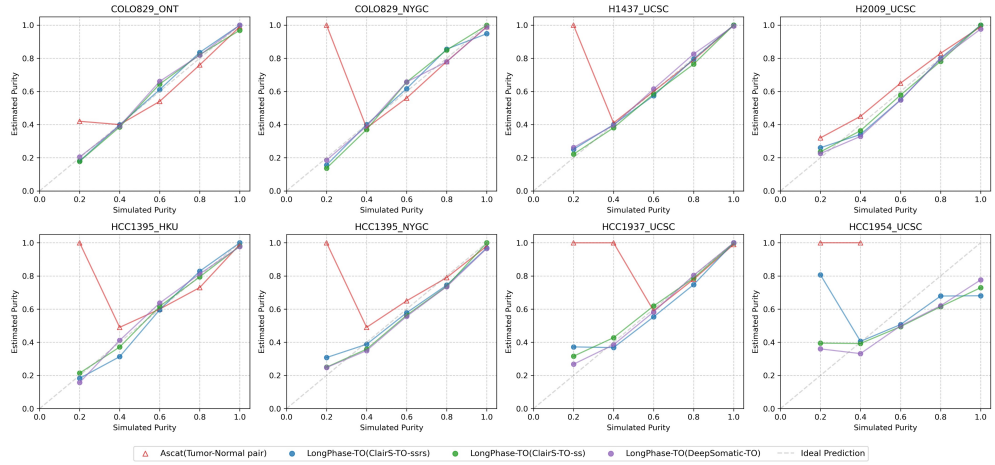


Fig. 7 Estimated tumor purity from three LongPhase-TO variants (LongPhase-TO(ClairS-TO-ssrs), LongPhase-TO(ClairS-TO-ss), and LongPhase-TO(DeepSomatic-TO)) and from ASCAT (tumor-normal paired analysis) was compared with simulated ground truth across eight cancer cell line datasets. Scatter plots display estimated purity (y-axis) against simulated purity (x-axis), with the diagonal line indicating the ideal prediction ($y = x$). LongPhase-TO pipelines showed high concordance with ground truth across purity levels (20–100%), with minor deviations at 20%. ASCAT exhibited greater variability, particularly at low purities.

ideal prediction line were observed at the lowest simulated purity (20%), reflecting the inherent challenges of analysis in low-purity samples, though overall performance remained favorable compared to alternative approaches.

In contrast, the ASCAT method exhibited reduced stability at low purity levels, with frequent deviations from the true values and occasional overestimation toward 100% purity. These findings indicate that the LongPhase-TO framework provides a generally accurate and reliable approach for tumor purity estimation from tumor-only long-read data, offering potential advantages over existing methods that require matched normal samples.

Qualitative Demonstration of Haplotype-Resolved Variant Analysis

To complement the quantitative benchmarks, we present qualitative evidence from genome browser visualizations that showcase the unique analytical power of our integrated pipeline. These examples illustrate the ability to resolve the haplotype context of individual somatic mutations and complex structural events.

Visualizing Haplotype-Resolved Somatic Variants

A key capability of our methodology is haplotagging, which refers to the assignment of a somatic variant to its parental chromosome of origin. A visualization from a cancer sample clearly demonstrates this process. In the displayed genomic region, long sequencing reads are successfully partitioned into two parental haplotypes (HP1 and HP2). A specific somatic variant is observed exclusively on a subset of reads belonging to Haplotype 2. These reads are further isolated into a distinct sub-haplotype track (HP2-1). This visualization provides unambiguous evidence that the somatic mutation occurred on the chromosome corresponding to Haplotype 2. Furthermore, its presence on only a subset of HP2 reads suggests the identification of a tumor subclone, highlighting the method’s potential to resolve intra-tumor heterogeneity (Figure 8).

Integrated Analysis of Co-occurring Somatic Alterations

Cancer genomes are often characterized by the co-occurrence of multiple types of alterations [38, 39]. A second, more complex visualization demonstrates the pipeline’s power to dissect such scenarios by simultaneously detecting and phasing a large-scale LOH event and a discrete somatic point mutation. In the displayed genome browser view (Figure 9), a large region on the left is clearly marked by a complete absence of sequencing reads assigned to Haplotype 1 (HP1), providing a definitive signature of LOH for that parental allele. To the right of this LOH region, the genome is heterozygous. Within this region, a specific somatic point variant is identified and precisely haplotagged. The variant is found exclusively on reads belonging to a sub-clonal population (HP2-1) of the retained Haplotype 2 (HP2).

This powerful example illustrates the capacity of the pipeline to reconstruct a complex mutational history: an LOH event resulted in the loss of the HP1 chromosome segment, and a subsequent somatic mutation arose on the remaining HP2 chromosome within a specific subclone. This ability to resolve different classes of somatic events and

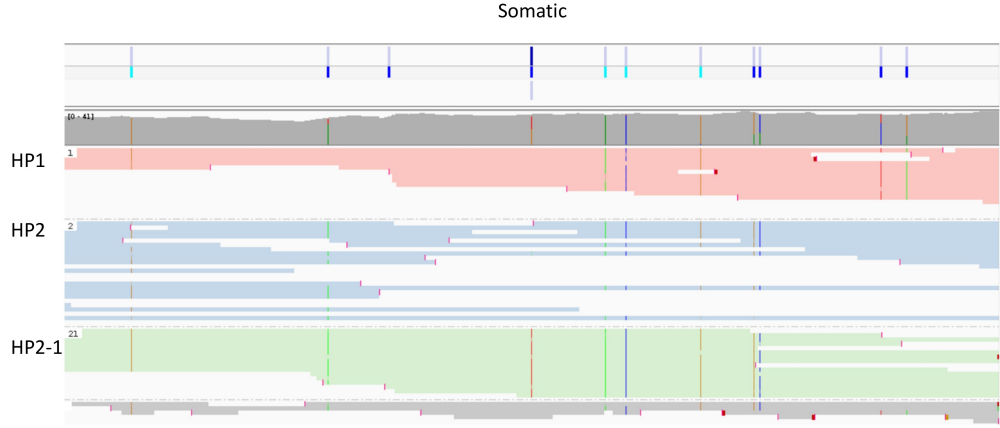


Fig. 8 Visualization of haplotype-resolved somatic variants from long-read sequencing. Genome browser snapshot illustrating the haplotype assignment (haplotagging) of a somatic variant. Long sequencing reads are partitioned into two parental haplotypes (HP1 and HP2). The somatic variant is detected exclusively on a subset of reads belonging to Haplotype 2, which are further separated into a sub-haplotype track (HP2-1). This provides direct evidence that the mutation originated from the chromosome corresponding to Haplotype 2. Moreover, its presence only on a fraction of HP2 reads indicates the detection of a tumor subclone, highlighting the capability of the method to resolve intra-tumor heterogeneity.

place them in their correct haplotype context is critical for a deeper understanding of tumor evolution and genetic architecture.

Conclusions

This thesis has addressed the growing need for specialized bioinformatics tools capable of leveraging the unique advantages of long-read sequencing for cancer genomics, particularly in clinical and research settings where only a tumor sample is available. The central contribution of this work is the development and validation of LongPhase-TO, a novel tumor-only analysis pipeline designed to provide a comprehensive, haplotype-resolved view of the somatic landscape. By integrating advanced phasing algorithms with graph-based modeling, this research demonstrates a new paradigm for extracting multi-layered genomic information from a single tumor specimen, thereby overcoming significant limitations of conventional methods.

The key findings of this study underscore the multifaceted capabilities of LongPhase-TO. First, the pipeline achieves accurate somatic phasing, enabling the assignment of somatic mutations to their specific parental chromosome of origin. This capability remains robust even in complex genomic regions, including those undergoing Loss of Heterozygosity (LOH), providing unprecedented resolution into the allelic context of carcinogenesis. Second, LongPhase-TO facilitates the reliable detection of large, chromosome-scale LOH events, with performance comparable to established array-based standards while avoiding the fragmentation issues that can plague other sequencing-based approaches. Third, by leveraging this phasing information, the pipeline significantly enhances the precision of somatic variant calling, a

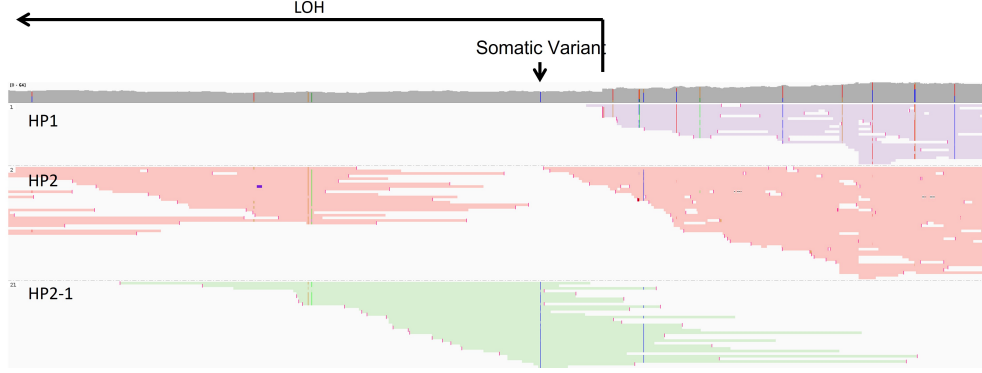


Fig. 9 Integrated visualization of LOH and haplotype-resolved somatic variant. Genome browser snapshot showing the simultaneous detection and phasing of a large-scale loss of heterozygosity (LOH) event and a somatic point mutation. The left genomic region is characterized by the complete absence of sequencing reads from Haplotype 1 (HP1), indicating LOH on this parental allele. To the right, the genome remains heterozygous. Within this region, a somatic point mutation is identified and haplotagged, appearing exclusively on a sub-clonal population (HP2-1) of the retained Haplotype 2 (HP2). This visualization reconstructs a complex mutational history, where LOH resulted in the loss of HP1 and a subsequent somatic mutation occurred on HP2 within a tumor subclone. Such integrative phasing provides critical insights into tumor evolution and the haplotype-specific architecture of somatic alterations.

benefit that is particularly pronounced in challenging cases characterized by low tumor purity. Finally, LongPhase-TO introduces a method for accurately estimating tumor purity directly from the sequencing data by analyzing signals of haplotype imbalance and LOH, thereby obviating the need for a matched normal sample and streamlining the analytical workflow.

The implications of this research are significant for both basic cancer biology and clinical oncology. For fundamental research, a haplotype-resolved analysis of the tumor genome offers deeper insights into the mechanisms of tumor evolution, including the specific allelic context of driver mutations and the clonal dynamics of LOH events. From a clinical perspective, the ability to perform robust, comprehensive genomic analysis from a tumor-only sample enhances the utility of long-read sequencing in diagnostic and prognostic settings where obtaining a matched normal sample is impractical or impossible. By delivering reliable LOH detection, improved variant calling, and accurate purity estimation within a single, integrated pipeline, LongPhase-TO is established as a cost-effective and broadly applicable tool for precision medicine.

While this work establishes a powerful new methodology, several avenues for future research can extend its capabilities and applications. Promising directions include the validation of LongPhase-TO across a broader spectrum of cancer types and a larger cohort of patient samples to further assess its generalizability and performance. The pipeline can also be extended to characterize other classes of complex structural variants, such as balanced translocations and chromothripsis, which are increasingly recognized for their role in cancer. Furthermore, integrating LongPhase-TO with other omics data, such as transcriptomics or epigenomics, could provide a more holistic

understanding of how genetic and epigenetic alterations cooperate in a haplotype-specific manner to drive tumorigenesis. Such advancements will continue to build upon the foundation laid by this thesis, further unlocking the potential of long-read sequencing to transform our understanding and treatment of cancer.

Methods

CNV and BFB Interval Calling

The identification of genomic intervals affected by large structural variants, such as CNVs and BFB-induced rearrangements, is a critical first step in characterizing the tumor genome. The method accomplishes this by systematically analyzing clipping patterns in long-read sequencing alignments. Clipping occurs when a portion of a sequencing read does not align to the reference genome, often because the read spans a structural variant breakpoint.

Principle of Clipping Pattern Analysis

The alignment of long reads across structural variant breakpoints generates characteristic and non-random patterns of clipping, including both soft and hard clipping. These patterns, encoded in the CIGAR (Compact Idiosyncratic Gapped Alignment Report) string of an alignment file, serve as informative signatures for specific SV types. A CIGAR string such as 10S5M2D1M indicates that the first 10 bases of the read were soft-clipped (10S), followed by 5 matching bases (5M), a 2-base deletion (2D), and one additional matching base (1M).

Distinct SVs produce unique spatial arrangements of these clipping types. For instance, a tandem duplication often presents as a sharp peak of up-clipped reads immediately followed by a sharp peak of down-clipped reads at the duplication boundaries. A fold-back inversion, a hallmark of BFB cycles, can produce a more complex signature of alternating and overlapping up-clipping and down-clipping signals. The algorithm is designed to computationally detect and delineate these signature-rich intervals.

Algorithmic Detection of Clipping Signatures

The formal identification of CNV and BFB intervals is performed using a three-step signal processing algorithm that transforms raw alignment data into defined genomic regions.

Clipping Breakpoint Extraction and Quantification

The algorithm first parses the alignment file (e.g., BAM format) for every clipped read. For each clipped read, the genomic coordinate of the clipping breakpoint is recorded. To create a directional signal, up-clipping events are assigned a positive value (e.g., +1) and down-clipping events are assigned a negative value (e.g., -1). This process converts the spatial distribution of clipped reads into a quantitative scatter plot of Clipping Count versus genomic position, where each point represents the net clipping signal at a single base (Figure 10).

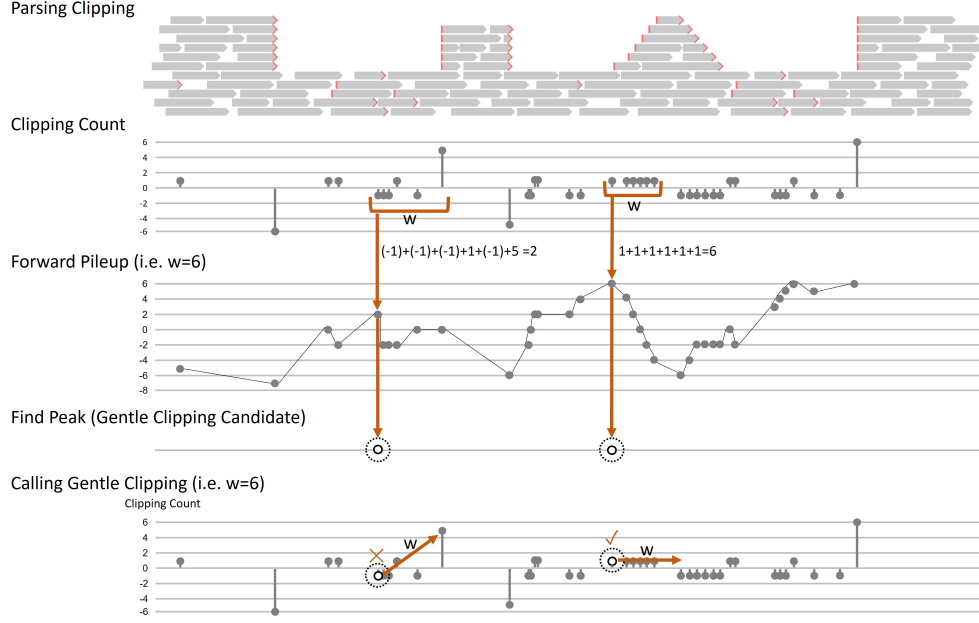


Fig. 10 The alignment file is parsed to extract clipped reads, and the breakpoint positions are transformed into directional clipping signals (Clipping Count). A sliding window summation (Forward Pileup, e.g., $w=6$) is applied to smooth the signal, enhancing consecutive low-amplitude events into discernible peaks. Peaks in the Forward Pileup profile are identified as candidate Gentle Clipping regions. To reduce false positives arising from isolated high-amplitude events, an additional filtering step is performed, ensuring that only regions characterized by sustained accumulative signals are retained as genuine Gentle Clipping events.

Gentle Clipping Signal Enhancement

Simple peak detection on the clipping count signal is insufficient to capture all relevant SV features. Therefore, an advanced feature detection strategy is introduced, specifically targeting gentle clipping, which refers to low amplitude but persistently occurring clipping events, and implemented through two consecutive steps for detection and signal quantification.

Gentle Clipping Boundary Identification. This criterion targets regions with a high density of consistent, low-amplitude clipping events rather than isolated high peaks. The signal is often sparse and noisy; to enhance true signals and reduce noise, detection begins with a smoothing step using a sliding window summation, termed Forward Pileup. For a window of size w , the pileup value is the sum of clipping counts within that window, e.g., $[1, 1, 1, 1, 1, 1]$ yields 6, while $[-1, -1, -1, 1, -1, 5]$ yields 2. This process transforms the discrete scatter plot into a continuous signal profile, amplifying sustained clipping events into distinct peaks and valleys (Figure 10).

The onset of a Gentle Clipping region can be located by identifying peaks in the Forward Pileup profile. Owing to its cumulative nature, a high pileup value indicates that subsequent positions predominantly contribute positive increments. When the signal transitions from sustained increase to decrease, the pileup value correspondingly

drops, making the peak a reliable marker for the start of a Gentle Clipping event (Figure 10).

However, a single high-amplitude clipping event within the window w can also yield a large Forward Pileup value, potentially mimicking the signal of Gentle Clipping. To prevent such false positives, an additional filtering step is applied to determine whether the elevated pileup value is caused by a single high peak or by a genuine accumulation of multiple low-level events. This distinction ensures that only sustained, high-density clipping regions are classified as Gentle Clipping candidates (Figure 10).

Similarly, the termination of a Gentle Clipping region can be identified by applying the same procedure in reverse. In this case, the Forward Pileup is replaced with a Reverse Pileup computation, and the peak detection logic is applied to points where the signal transitions from a positive to a negative trend. This corresponds to regions where the cumulative value exhibits a sustained decrease and reaches a local minimum, marking the boundary at which the Gentle Clipping event ends. This symmetric detection framework ensures that both the onset and termination of Gentle Clipping regions are delineated in a consistent and accurate manner.

Gentle Clipping Amplitude Quantification. To improve the interpretability of identified Gentle Clipping sites, an amplification step that quantifies the relative magnitude of local signal variation. This process begins by constructing a Clipping Pileup profile, derived from summing the raw Clipping Count values for each genomic coordinate, thereby preserving both the directionality and magnitude of clipping events.

For each Gentle Clipping candidate, two reference points are defined: one located w bases upstream and the other w bases downstream of the candidate position, where w corresponds to the predefined window size used in the Forward Pileup procedure. The Clipping Pileup values at these two positions are then retrieved, and their difference is calculated to quantify the local amplitude of change (Figure 11).

A higher amplitude value denotes a more pronounced and abrupt transition in the clipping signal, indicative of a stronger and more distinct breakpoint signature. This quantitative metric is retained for downstream analyses to guide the identification of breakpoint-associated genomic intervals.

Breakpoint-Guided Interval Calling

Following the Advanced Signal Feature Detection step, the signal amplitudes from the Gentle Clipping process are further enhanced, ensuring that low-intensity fluctuations no longer contain features of interest. Consequently, signal components with amplitudes below the threshold λ can be confidently discarded, while retaining only the prominent signal peaks that represent meaningful biological events.

Following this noise reduction, the core detection mechanism relies on the concept of Signal Swing, which is defined by two complementary conditions that assess both signal similarity and spatial proximity. For a pair of signal points i and j with clipping pileup values c_i and c_j respectively, a signal swing is identified when these points satisfy the following criteria:

$$(c_i/c_j \geq \alpha) \wedge (|j - i| \leq \beta)$$

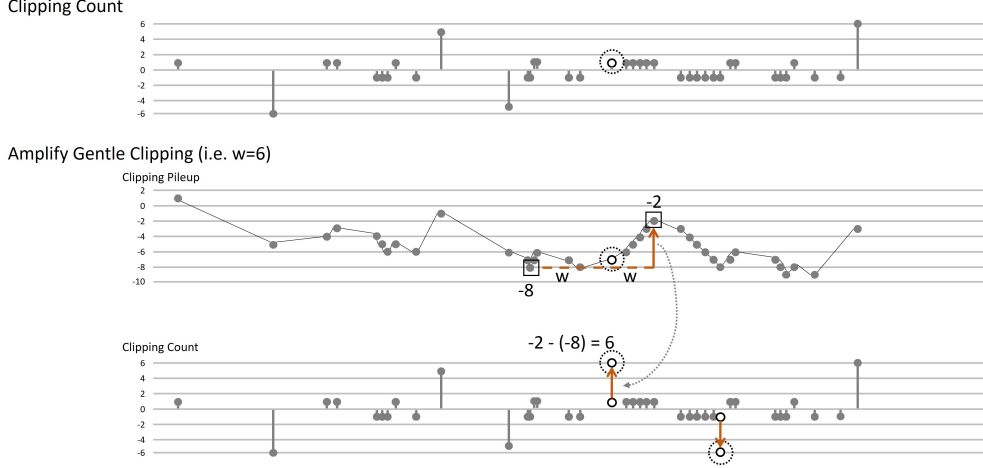


Fig. 11 Illustration of the amplification procedure for Gentle Clipping signal quantification. The raw clipping counts are first aggregated into a Clipping Pileup profile, which retains both the directionality and magnitude of clipping events. For each Gentle Clipping candidate, two reference points are selected at positions w bases upstream and downstream, where w corresponds to the predefined sliding window size. The Clipping Pileup values at these reference positions are compared, and their difference is computed to measure the local signal amplitude, thereby improving the interpretability of Gentle Clipping sites.

The first condition, $c_i/c_j \geq \alpha$, evaluates the ratio of signal amplitudes to determine whether the two points exhibit similar signal characteristics, where α represents the minimum amplitude ratio threshold. The second condition, $|j - i| \leq \beta$, ensures spatial proximity by constraining the maximum genomic distance between the signal points to β base pairs.

The genomic coordinates derived from validated signal swings serve as the boundaries for distinguishing between larger genomic intervals and smaller regions, enabling precise identification of CNV and BFB candidate regions (Figure 12).

Chromosome-Scale Loss of Heterozygosity Detection

Loss of Heterozygosity (LOH) is a hallmark of cancer genomes, reflecting the loss of one parental allele over a large genomic region. LOH events are detected at a chromosome scale by integrating variant information with the previously identified CNV and BFB intervals.

LOH Detection via Heterozygosity Ratio

The primary signal for LOH is a stark imbalance in allelic ratios at heterozygous variant sites. For each known heterozygous single nucleotide variant (SNV), a heterozygosity ratio is calculated based on the read counts supporting the heterozygous versus homozygous state. This is defined as:

$$\text{Heterozygosity ratio} = \frac{\text{Het Count}}{\text{Het Count} + \text{Hom Count}}$$

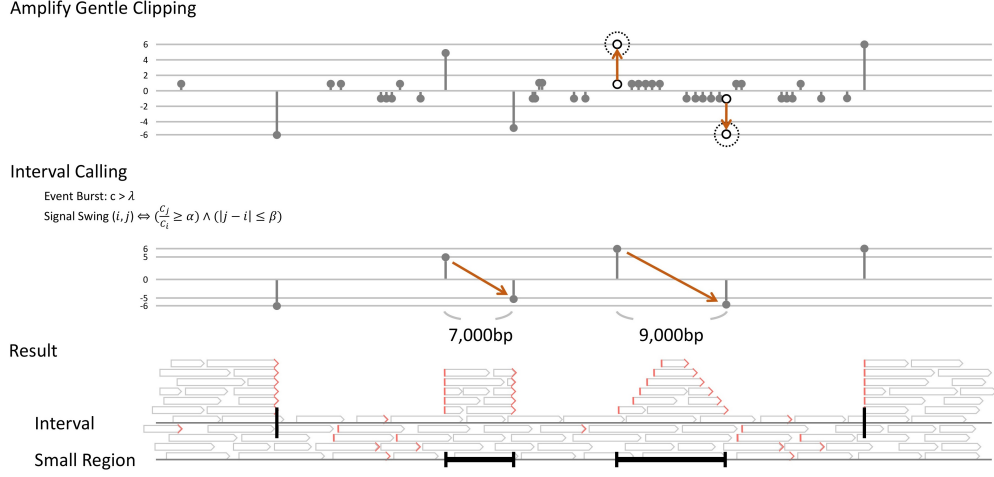


Fig. 12 Only prominent clipping signal peaks are retained after signal enhancement, and genomic intervals are subsequently identified based on combined constraints of signal amplitude and spatial distance. The resulting genomic coordinates define the boundaries of intervals and sub-regions, enabling precise detection of candidate CNV and BFB regions.

In a normal diploid region, this ratio is expected to be close to 1.0 (or a high value, accounting for mapping biases). In a region of LOH, where one allele is lost, the vast majority of reads will support the remaining allele, causing the heterozygosity ratio to drop to a value near zero. An LOH event is called if this ratio falls below a predefined threshold, σ :

$$\text{Heterozygosity ratio} < \sigma$$

Visually, this corresponds to the complete disappearance of long reads supporting one of the two haplotypes over a large genomic span.

Merging LOH Segments for chromosome-Scale Analysis

A key challenge in LOH detection is that large LOH regions can be interrupted by small, localized genomic events such as tandem duplications (CNVs) or complex rearrangements from BFB cycles. These events can locally restore a heterozygous state or create complex copy number profiles, causing transient spikes in the heterozygosity ratio and fragmenting the LOH call.

To accurately identify true, contiguous chromosome-scale LOH events under these circumstances, this method incorporates a critical filtering and merging step that explicitly leverages CNV and BFB intervals obtained in the previous stage. Without integrating these previously identified regions, large LOH segments could be fragmented by such small, localized events, making it impossible to recover the complete chromosome-scale pattern. During computation, the algorithm deliberately skips over these small, intervening genomic events to preserve the major, uninterrupted LOH span (Figure 13), enabling the reconstruction of a single, continuous LOH event that provides a more biologically accurate representation of a major chromosome alteration than treating it as multiple disjointed smaller events.

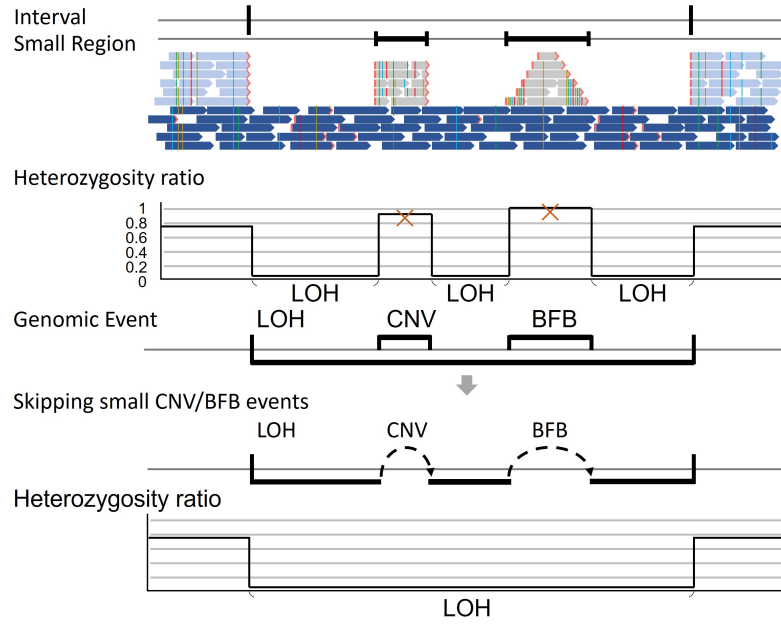


Fig. 13 Method for chromosome-scale LOH detection. The workflow shows that an initial pass may identify fragmented LOH regions interrupted by small CNV or BFB events that locally restore heterozygosity. The final step involves a merging process that filters out these small intervening events to define a single, continuous, large-scale LOH region.

Somatic Variant Identification and Phasing

The accurate identification of somatic variants from tumor-only long-read data requires a multi-stage process that combines rigorous filtering, advanced haplotype-aware variant calling, and high-resolution phasing.

Somatic Variant Candidate Selection

The initial list of variant candidates produced by a standard caller is a heterogeneous mixture of true somatic mutations, common germline variants, and technical artifacts. The first step is to enrich this list for true somatic events.

Filtering Against a Panel of Normals (PON)

In the absence of a matched normal sample, a Panel of Normals (PON) is used to filter out common germline variants. The PON is a comprehensive database constructed from a large cohort of healthy individuals (e.g., 1000 Genomes Project, gnomAD) and also includes recurrent, technology-specific artifacts. All variant candidates from the tumor sample that are present in the PON are removed in a subtractive process that effectively eliminates the majority of non-somatic variants (Figure 14).

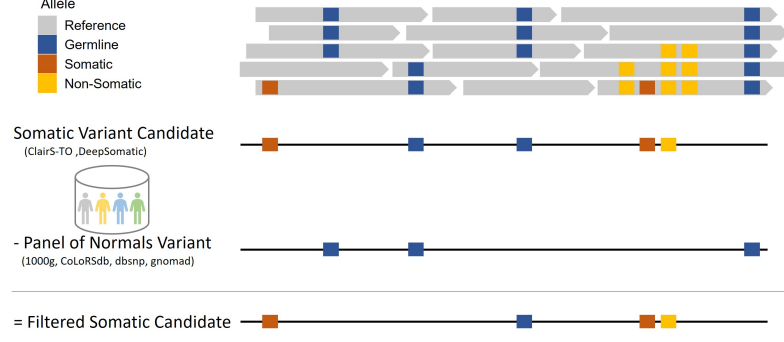


Fig. 14 Filtering somatic variant candidates using a Panel of Normals (PON). This schematic shows that an initial set of variant candidates, containing a mix of somatic (brown), germline (blue), and other variants, is filtered by subtracting all variants found in the PON. This process enriches the final candidate list for true somatic mutations by removing common germline polymorphisms and known artifacts.

Filtering of Clustered Variants via Allele Concordance

Systematic sequencing and alignment errors often manifest as clusters of false positive variants in close proximity. To identify and remove these artifacts, an Allele Concordance filter is employed. This method evaluates a candidate variant at position i by analyzing its co-occurrence with neighboring variants on the same long reads. For a candidate variant i and a nearby variant j , two concordance ratios are calculated:

$$\frac{AltAlt_{ij}}{AltAlt_{ij} + AltRef_{ij}} \geq \mu$$

$$\frac{AltAlt_{ij}}{AltAlt_{ij} + RefAlt_{ij}} \geq \nu$$

where $AltAlt_{ij}$ is the number of reads with alternate alleles at both i and j , $AltRef_{ij}$ is the count of reads with an alternate allele at i and reference at j , and $RefAlt_{ij}$ is the count of reads with reference at i and alternate at j . The thresholds μ and ν control the stringency. If a candidate variant i shows high concordance with multiple neighbors (i.e., its aggregate Allele Concordance score exceeds a final threshold ρ), it is classified as a clustered artifact and removed from the candidate list (Figure 15).

Graph-Based Somatic Variant Calling

The filtered list of high-confidence candidates is then subjected to a novel, graph-based variant calling algorithm designed to leverage local haplotype information.

The Tri-nodal-edge Graph Model

Since somatic variants arise post-zygotically and humans are diploid, each somatic mutation must originate from one of the existing haplotypes and subsequently diverge from it. In contrast, sequencing artifacts are expected to occur randomly and therefore lack such haplotype-specific origins. Based on this principle, the Tri-nodal-edge

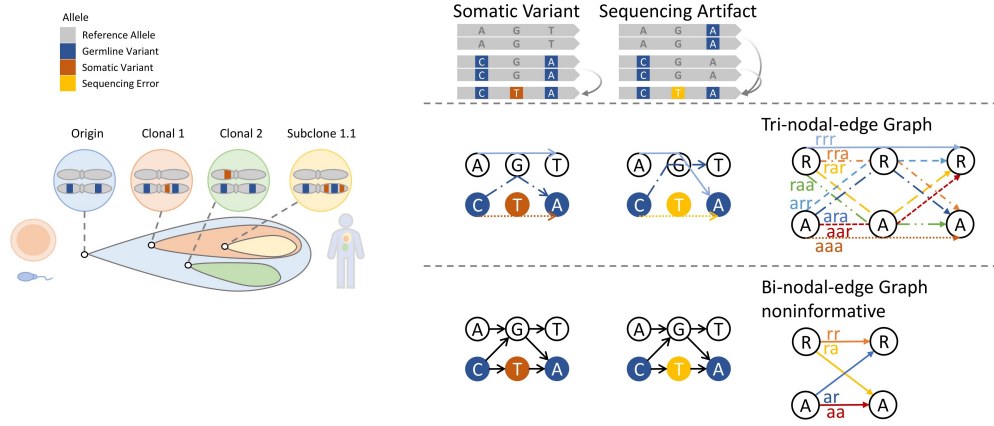


Fig. 16 Comparison of graph models for somatic variant calling. The figure contrasts the proposed Tri-nodal-edge Graph, which models haplotypes across three adjacent variant sites, with a simpler Bi-nodal-edge Graph that only considers pairs. The Tri-nodal-edge graph captures more linkage information, making it more powerful for distinguishing true somatic variants from artifacts, while the Bi-nodal-edge graph is labeled noninformative for this complex task.

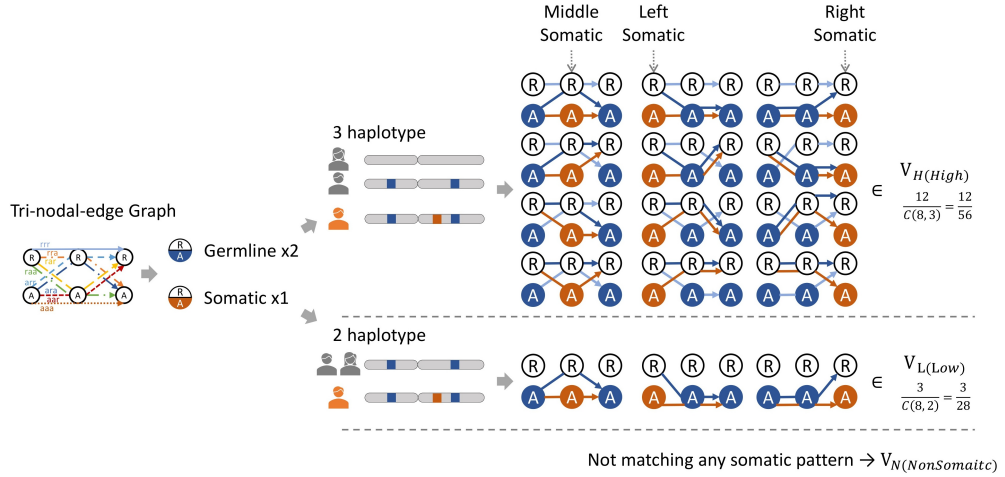


Fig. 17 High-confidence patterns (V_H) arise from three-haplotype scenarios with two distinct germline haplotypes and one somatic haplotype, while low-confidence patterns (V_L) arise from two-haplotype scenarios with identical germline haplotypes and one somatic haplotype. All other combinations are classified as non-somatic (V_N).

possible 2-haplotype combinations, the study identifies 3 specific configurations that are classified as Low-Confidence Somatic V_L patterns.

- **Non-Somatic V_N :** Any observed haplotype combination not matching a V_H or V_L pattern.

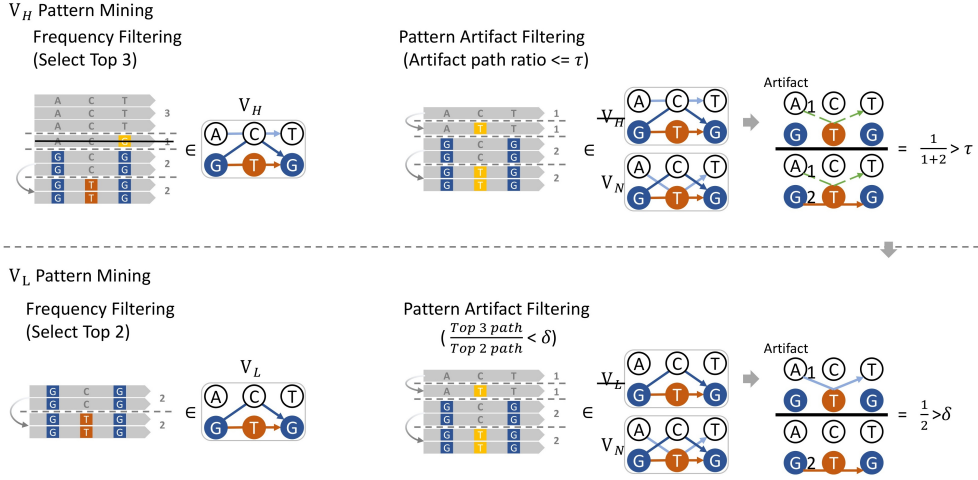


Fig. 18 Filtering strategy for V_H and V_L haplotype patterns. Top-ranked haplotype paths are first selected by read support, followed by artifact-based reclassification. Variants exceeding the noise thresholds (τ for V_H , δ for V_L) are reassigned as V_N , ensuring accurate somatic variant detection in complex haplotype contexts.

Pattern Mining and Artifact Filtering

In practical applications, the haplotype graph often contains more than three possible paths between alleles. To determine whether a candidate variant conforms to a V_H (High-Confidence Somatic) pattern, all possible paths (up to eight in total) by their supporting read counts and select the top three for evaluation. This top-3 selection increases sensitivity but also introduces the possibility of ties in path frequencies. Consequently, a candidate variant may simultaneously match both V_H and V_N (Non-Somatic) pattern criteria.

To address such ambiguity, an artifact path ratio metric, which quantifies the proportion of sequencing noise within the top-3 paths that match the V_H pattern definition. Specifically, if the artifact path ratio exceeds a predefined threshold τ , the variant is reclassified from V_H to V_N , thereby mitigating false-positive somatic calls due to sequencing artifacts.

A similar strategy is applied for V_L (Low-Confidence Somatic) pattern detection. In this case, only the top two most frequent paths are selected for evaluation. As with V_H , ties in path frequencies can lead to ambiguous classification between V_L and V_N . To resolve this, the relative artifact proportion between the top-3 and top-2 path sets. If the noise contribution within the V_L pattern exceeds a separate threshold δ , the candidate is downgraded to V_N (Figure 18).

This two-tiered filtering approach—frequency-based path selection followed by artifact-based reclassification—ensures that both V_H and V_L detections maintain high precision while retaining sensitivity in complex haplotype contexts.

Final Somatic Decision Formula

In practice, the decision for a candidate variant is not made from a single local haplotype graph, but rather from the combined evidence of multiple graphs spanning a

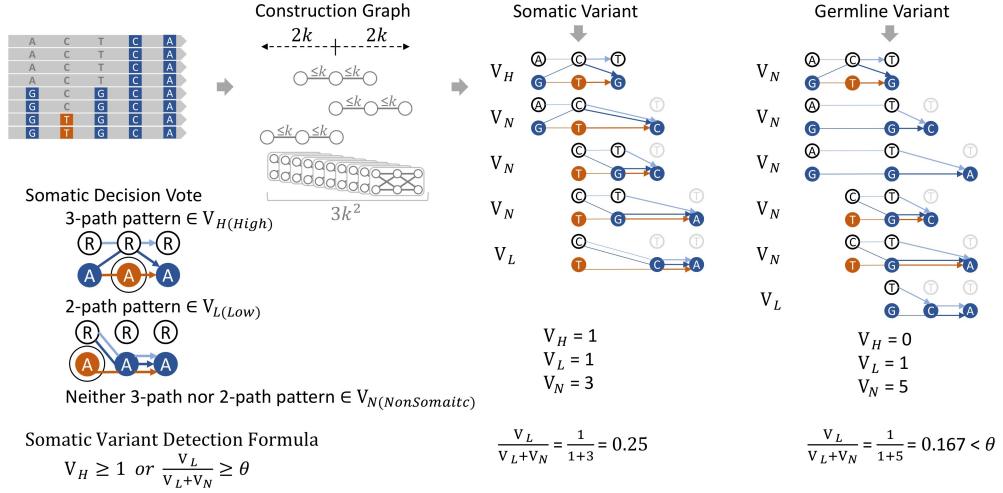


Fig. 19 The final decision-making process for somatic variant calling. This diagram illustrates how all local haplotype patterns at a site are classified as high-confidence V_H , low-confidence V_L , or non-somatic V_N . These classifications are tallied in a voting system, and a final decision is made based on a formula that prioritizes V_H evidence or a significant proportion of V_L evidence.

broader genomic context to capture richer haplotype linkage information. Specifically, for a site under evaluation, k variants (where k denotes the number of variant sites) to both the upstream and downstream directions, resulting in a total span of $2k$ bases around the focal point. Within this window, overlapping subgraphs are constructed, such that each local graph covers the central site and the k surrounding positions on either side. This process yields k^2 distinct Tri-nodal-edge graphs on each side of the focal site, with possible overlap between the left and right contexts. Consequently, a total of $3k^2$ local graphs contribute to the classification decision for a single site.

The final somatic determination is made using a voting-based aggregation system, which integrates the evidence from all local haplotype patterns (Figure 19). A site is classified as somatic if it satisfies either of the following conditions:

$$V_H \geq 1 \quad \text{or} \quad \frac{V_L}{V_L + V_N} \geq \theta$$

This hierarchical rule ensures high sensitivity by calling any site with strong V_H evidence, while maintaining specificity by requiring weaker V_L evidence to surpass a statistical threshold θ .

LOH-Aware Somatic Phasing

To resolve haplotype structures from long-read sequencing data, we constructed a directed acyclic graph (DAG) in which each heterozygous variant V_i is represented by two vertices corresponding to its reference and alternative alleles, v_i and v'_i . Edges in the graph encode the co-occurrence of allele pairs observed within individual long sequencing reads. For two adjacent variants V_i and V_{i+1} , four possible directed edges— (v_i, v_{i+1}) , (v_i, v'_{i+1}) , (v'_i, v_{i+1}) , and (v'_i, v'_{i+1}) —are assigned weights corresponding to

the number of reads supporting each allele pair, thereby quantifying the strength of evidence linking alleles across variant positions.

Haplotype reconstruction begins at the leftmost variant, assigning allele v_i to haplotype 1 (HP1) and allele v'_i to haplotype 2 (HP2). Using these seeds, haplotypes are extended iteratively from left to right. Because genomic regions differ in zygosity and somatic status, the extension procedure is performed under three scenarios: non-somatic regions, somatic sites, and transitions between runs of homozygosity (ROH) and loss of heterozygosity (LOH).

In non-somatic regions, two independent allele paths are considered. Haplotype assignment at V_{i+1} relies on comparing the total evidence linking consistent versus inconsistent allele combinations across the two haplotypes. Specifically, we evaluate the combined support for paths (v_i, v_{i+1}) together with (v'_i, v'_{i+1}) against the support for (v_i, v'_{i+1}) combined with (v'_i, v_{i+1}) . If the former exceeds the latter, allele v_{i+1} inherits the haplotype of v_i ; otherwise, the assignment is inverted.

At somatic variant positions, the graph structure differs because one allele is shared across paths while the somatic allele forms a branching structure. Assuming v'_{i+1} is the somatic allele, the non-somatic allele v_{i+1} becomes uninformative for haplotyping. Haplotype assignment is therefore determined by comparing the read support for the two allele pairs involving the somatic allele: (v'_i, v'_{i+1}) versus (v_i, v'_{i+1}) . The somatic allele is assigned to the haplotype associated with the stronger of these two edges.

In ROH–LOH transition regions, one allele is absent, reducing the local graph to a single path. If v'_{i+1} lies within an LOH region, allele v_{i+1} is not observed, and haplotype assignment relies on comparing evidence from (v_i, v'_{i+1}) versus (v'_i, v'_{i+1}) . The allele follows whichever of these two edges carries greater support.

Both heterozygous and homozygous variants are incorporated within this unified graph-based framework. Homozygous variants naturally conform to the simplified LOH-like graph topology, and their haplotypes are inferred by propagating assignments from flanking regions. Heterozygous variants follow the inference rules described above for non-somatic and somatic scenarios. Importantly, because long reads often span many variants, we generalize the decision-making process to incorporate multi-variant (k -variant) co-occurrence evidence whenever available. Integrating support across multiple linked variants improves robustness to local noise, sequencing artifacts, and fluctuations in variant allele frequency, thereby enhancing overall phasing accuracy.

Tumor Purity Prediction

Accurate estimation of tumor purity—the fraction of cancer cells in a sample—is essential for the correct interpretation of somatic variant data. Purity is predicted by modeling the genomic consequences of mixing tumor and normal cell populations.

Principle of Haplotype Imbalance

Information derived from haplotype imbalance (HI) can be further leveraged to estimate tumor purity. In practice, sequencing data often represent a mixture of tumor and normal cell populations. In a purely normal sample, the two germline haplotypes

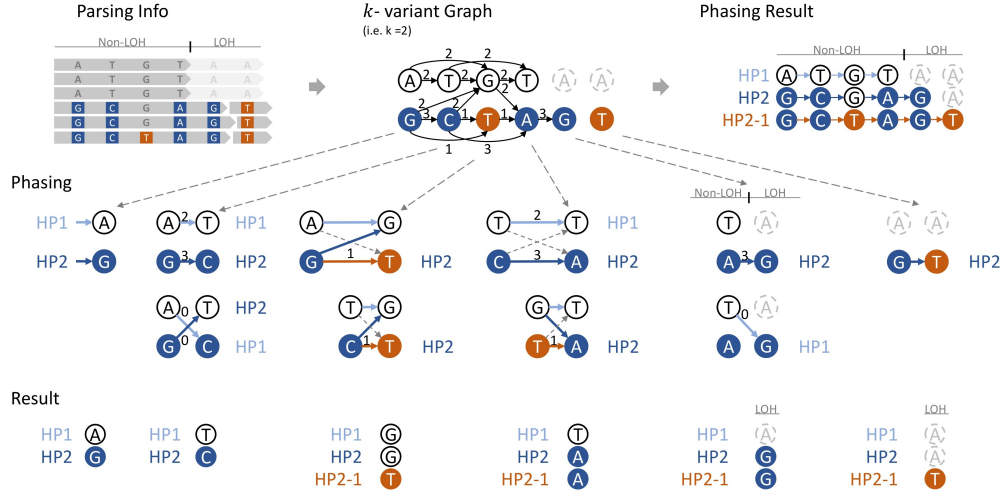


Fig. 20 Graph-based haplotype phasing workflow. This diagram illustrates how sequencing read data is transformed into a k-variant graph, where nodes are alleles and weighted edges represent their co-occurrence on reads. The algorithm then traverses this graph to reconstruct the most likely haplotypes, including distinguishing between germline (HP1, HP2) and newly evolved somatic (e.g., HP2-1) haplotypes.

are expected to be equally represented, resulting in balanced HI values close to 0.5. As the fraction of tumor cells increases, this balance is progressively disrupted, and HI values shift away from 0.5 toward 1.0, reflecting increasing haplotype imbalance.

Formally, HI at each heterozygous site is defined as:

$$HI = \frac{\max(\text{Count}_{HP1}, \text{Count}_{HP2})}{\text{Count}_{HP1} + \text{Count}_{HP2}}$$

where Count_{HP1} and Count_{HP2} represent the allele-specific read counts assigned to the two germline haplotypes. To quantify the relationship between HI and tumor purity, HI is computed across all somatic variant sites and subsequently aggregated to capture the genome-wide distribution. Visualization using histograms or boxplots reveals a characteristic pattern: samples with low tumor purity exhibit HI values tightly centered around 0.5, whereas samples with high tumor purity display a systematic shift toward 1.0. This distributional change provides a quantitative basis for purity modeling. Accordingly, polynomial regression is applied to the observed HI distributions to establish a predictive relationship between HI and tumor purity. Once the regression model is trained, HI values derived from sequencing data can be directly input into the fitted function to yield accurate estimates of tumor purity (Figure 21).

Predictive Modeling with Genomic Features

To formalize this relationship, a polynomial regression framework was applied. Two key sets of features were extracted from the genomic data. The first was the LOH Ratio, defined as the proportion of the genome affected by LOH:

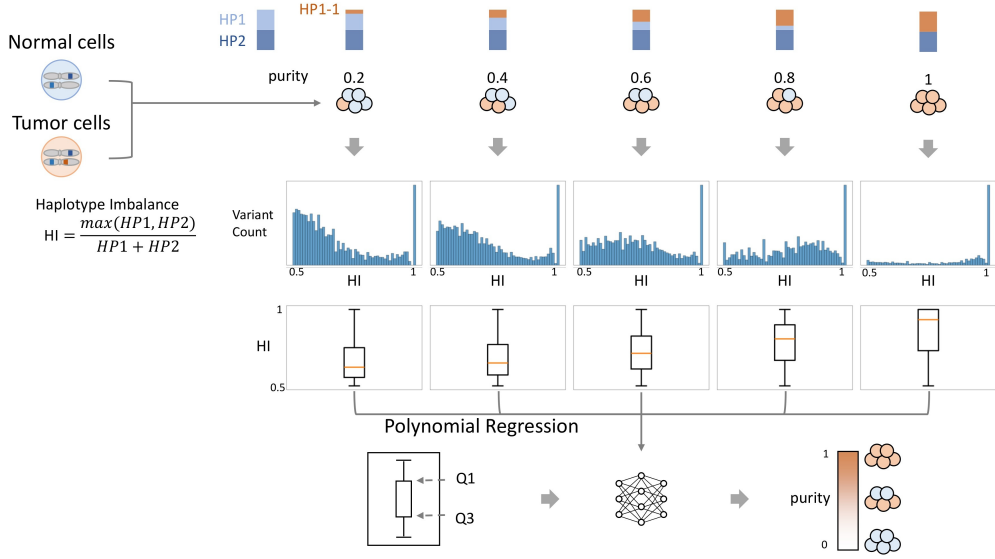


Fig. 21 Principle of tumor purity prediction using Haplotype Imbalance (HI). The figure shows that as tumor purity increases from 0.2 to 1.0, the proportion of somatic haplotypes increases. This causes the genome-wide distribution of HI, shown in histograms and box plots, to shift from a balanced state (centered at 0.5) to a highly imbalanced state with higher median values and a peak at 1.0.

$$\text{LOH Ratio} = \frac{L_{\text{LOH}}}{L_{\text{Genome}}}$$

where L_{LOH} denotes the total length of all detected LOH regions and L_{Genome} is the total analyzable genome length. Although the regression model primarily relied on haplotype imbalance (HI) as the main predictor, the LOH ratio provided complementary information. In particular, LOH is especially pronounced when tumor purity approaches 1, thereby improving predictive accuracy in high-purity samples.

The second feature set comprised the haplotype imbalance statistics, namely the first (Q1) and third (Q3) quartiles of the genome-wide HI distribution, which summarize the central tendency and variability of allelic imbalance. Together, these features (LOH Ratio, Q1 of HI, and Q3 of HI) were used as predictors in the polynomial regression model, which was fitted to establish the relationship between genomic features and tumor purity (Figure 22).

Data and Computational Resources

Data Sources and Benchmark Sets

This study made use of publicly available long-read sequencing data derived from a panel of well-characterized cancer cell lines. The inclusion of these standardized reference materials ensures the reproducibility of the analyses and enables rigorous benchmarking against established ground-truth call sets. The cell lines, together with their corresponding data sources and benchmark sets, are summarized in Table 1.

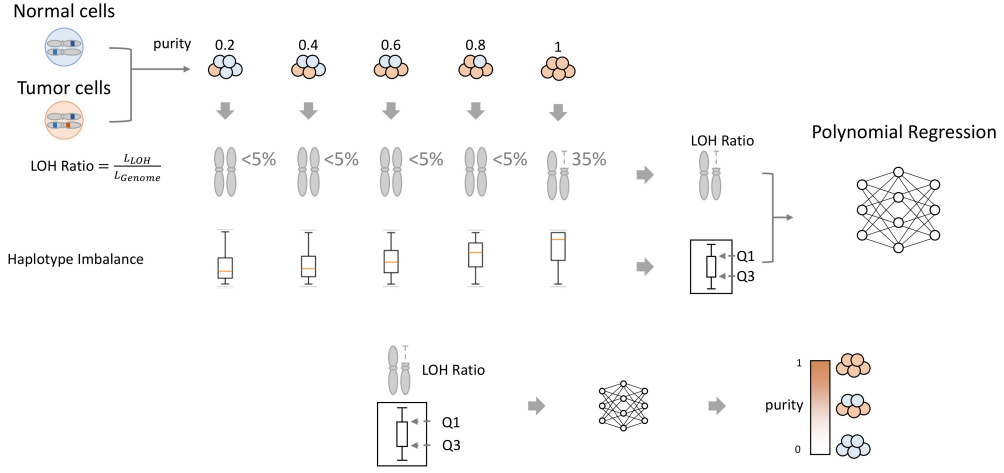


Fig. 22 Schematic of the tumor purity prediction model. The model takes two types of genomic features as input: the LOH Ratio (fraction of the genome with LOH) and statistics from the Haplotype Imbalance distribution (represented by the Q1 and Q3 quartiles of a box plot). These features are provided to a Polynomial Regression model, from which the estimated tumor purity is obtained.

Table 1: Cancer cell lines used in this study, with their sequencing data sources and benchmark references.

Cell Line	Material Source	Benchmark Source
COLO829	ONT, NYGC	NYGC
H1437	UCSC	Google
H2009	UCSC	Google
HCC1395	HKU, NYGC	SEQC2
HCC1937	UCSC	Google
HCC1954	UCSC	Google

Somatic Variant Calling Tools

To assess the performance of the proposed pipeline, we compared it against several state-of-the-art somatic variant calling tools specifically designed for long-read sequencing and/or tumor-only analysis. The tools and their respective versions are summarized in Table 2.

Table 2: Somatic variant callers used for benchmarking.

Somatic Variant Callers
ClairS-TO v0.3.0 (ssrs)
ClairS-TO v0.3.0 (ss)
DeepSomatic v1.8.0 (tumor-only)

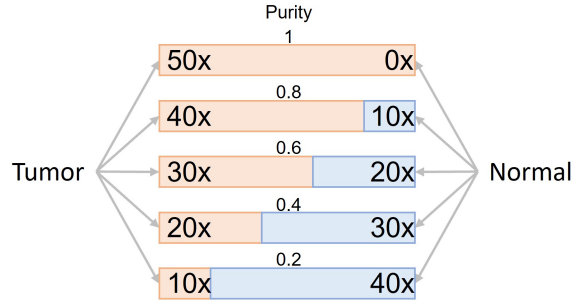


Fig. 23 Generation of in-silico tumor–normal mixtures for benchmarking. This diagram illustrates the method of creating synthetic samples with known tumor purity. Raw sequencing reads from a pure tumor sample (orange) and a pure normal sample (blue) are mixed in specific proportions to create samples with precise purity levels (e.g., 0.8 purity from 40x tumor and 10x normal reads), providing a ground-truth dataset for model training and validation.

Generation of In-Silico tumor–normal Mixtures

To test and evaluate the accuracy of variant calling under different tumor purities, and to train and validate the tumor purity prediction model, a benchmark dataset was required, with purity levels assumed to be known (assuming the pure tumor cell line has purity = 1). This dataset was generated using an in-silico titration approach. High-coverage long-read sequencing data were obtained from a pure tumor cell line and its matched normal cell line. Synthetic mixture samples were then created by downsampling the raw reads and mixing tumor and normal reads at predefined ratios under different total coverage levels (Figure 23). Specifically, five target purity levels were set: 1.0, 0.8, 0.6, 0.4, and 0.2, and corresponding mixed samples were generated at different total coverages. This procedure resulted in a benchmark dataset, where each sample was assigned a corresponding purity label, which can be used to assess the performance of variant calling and serve as a reference for training and validating tumor purity prediction models.

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